A fixed first-class plan for the prefix counter

Human Compression project

We acquire the accepted prefix enumeration, first-class partition proof, and version-two receipt admission. Different executions of the group library may return different ordered representatives of the same complete conjugacy classes. Version two correctly rejects their literal union. The present caller instead takes one explicit, hash-bound list and validates it against the actual group on every invocation. No random seed or equality of class indices is assumed.

Let G be the fixed literal finite permutation group. A proposed plan is an ordered list $(a_1, \dots, a_s)$ of permutations on its actual moved-point domain. Independently compute the complete conjugacy classes of G, and let $(b_1, \dots, b_t)$ be their representatives of order greater than two. The completeness of this library interface is the same retained software premise as in the accepted counter.

Proposition 1. *Suppose $s = t$, every a_i belongs to G and has order greater than two, and there is a permutation π of $\{1, \dots, s\}$ and elements $g_i \in G$ satisfying $a_i^{g_i} = b_{\pi(i)}$. Then the fixed list contains exactly one representative of every conjugacy class of order greater than two. Executing only branch i of the accepted counter from $(1, a_i)$, and joining exactly the branches $1, \dots, s$, therefore enumerates exactly the same inner marking orbits as the accepted whole counter.*

Proof. Surjectivity of π places every required conjugacy class in the fixed list. If two fixed entries were conjugate in G, their corresponding b entries would be conjugate and hence represent the same library class, contradicting injectivity of π. The fixed entries therefore are a complete transversal. The acquired partition proof applies to any such transversal: every marking has its first entry in exactly one of these classes, while subsequent prefix centralizers and intersections are computed by the unchanged recursion. Conjugating a marking transports it to the selected representative. Consequently different first branches cannot duplicate an inner orbit and every orbit occurs. Earlier results and counters are not read when choosing a recursive extension. □

The helper checks actual membership and order. For each fixed entry it finds an actual conjugating element to one fresh representative, checks that element's membership in G and the conjugacy equality, and rejects any reused class. Equal list lengths and the checked bijection prove coverage. The returned certificate stores the two full literal lists, the bijection and all conjugating permutations. The Python admission reader independently verifies the permutation domains, bijection and literal conjugacy identities; group membership and fresh-class completeness are acquired from the exact executed helper. A proposal extracted from an older census file is only a proposal: its prior count or claimed class indices confer no validity.

The only scientific code changes are this helper, two added arguments, and replacement of the original first-list assignment by the fixed list and its validation. The complete recursive visit, subsequent outer loop, marking tests, automorphism computation and result writer are byte-identical to version two. The full involution list still comes from the fresh complete conjugacy classes.

Receipt boundary. Every invocation requires the exact new operational review, with its exported dependency file binding this author edition. All earlier manifest/export/dependency checks remain. The plan and model are mandatory pinned inputs. The request and driver are reconstructed from their exact bytes; the driver loads the actual new counter and embeds the full fixed list. The collector requires the precise command, clean terminal status, complete unchanged

input and output inventories, and the returned plan certificate. Every census must contain the entire pinned list and start at the requested representative. A union requires one identical literal model, plan, engine, author and operational acceptance, and exactly every index once. An external deadline grants no branch credit. Neither a partial class list nor a differently located plan is silently substituted. Division by the outer automorphism index is permitted only after this complete union, exactly as before.

Evidence and scope. The retained version-two L3(4) extension calls returned with differing literal class lists; their rejected union and complete individual files remain historical evidence, not a new union certificate. Small actual A_4 controls compare two complete fixed branches with the acquired whole result. Separate actual group checks reverse and conjugate the backend representatives while keeping the fixed list unchanged, and also reverse the proposed plan. Duplicated, missing, extra, identity, involution and outside-group entries are rejected. Receipt and gate damages preserve the distinction between explicit test-only fixtures and production admission. No source-group branch is executed by these checks.

The original partition theory and counter were authored by the parent agent; this fixed-plan helper and operational successor were authored by the reviewer of the former receipt interface, with distinct review of this new work reserved to the parent. No source count, mathematical classification, whole-source completion or performance improvement is claimed by this caller alone.